FINAL PROJECT REPORT

Project Title: Airfare Price Prediction Using Machine Learning

Group Number: 09

Team Leader: Sachin Anthony

Team Members: Sachin S, Deepak N B, Alisha Verma, Karan Demta, Chandru BS

Mentor: Aishwarya Sarda

Date: January 18, 2026

1. Summary of Problem Statement, Data, and Findings

Abstract The U.S. aviation market is characterized by opaque, dynamic pricing algorithms that often leave travelers overpaying for flights. This project aimed to develop a transparent, data-driven system to predict the "Fair Market Price" of domestic airfares. Utilizing the **U.S. Consumer Airfare Report (Table 1a)**, we analyzed over 200,000 route records. Our initial hypothesis—that distance is the primary cost driver—was disproven by a baseline Linear Regression model (58% accuracy).

Key Findings:

- **Location is King:** Implementing **Target Encoding** for specific City-Pair routes (e.g., "JFK-LAX") was the single most critical factor, boosting model accuracy from 68% to **82.34%**.
- **Monopoly Impact:** We statistically confirmed that routes with high **"Monopoly Intensity"** (Market Share * Distance) carry a significant price premium.
- **Outcome:** The final XGBoost model achieved a Mean Absolute Error (MAE) of **$19.36**, meaning our predictions are accurate within ~$20 of the real ticket price. This engine was successfully deployed via a Flask web application ("SkyCast AI") with an integrated chatbot.

2. Overview of the Final Process

Our methodology followed the **CRISP-DM** (Cross-Industry Standard Process for Data Mining) framework:

1. **Data Acquisition:** Merged the core BTS Table 1a dataset (Quantitative) with scraped Skytrax Reviews (Qualitative) to capture both market fundamentals and passenger sentiment.
2. **Preprocessing:** Applied rigorous cleaning to handle missing market share data and removed "Leakage" variables (like `fare_lg`) that would artificially inflate accuracy.
3. **Feature Engineering:** Created novel interaction terms such as `dominance_gap` (Difference between dominant and low-cost carrier share) to quantify market power.

4. **Modeling Strategy:** Moved from a baseline **Linear Regression** (to test linearity) to tree-based ensembles (**Random Forest**, **Gradient Boosting**), culminating in an optimized **XGBoost** regressor.
5. **Deployment:** The "Champion Model" was serialized (`.pkl`) and integrated into a **Flask** web application with a responsive UI and an NLP-powered chatbot.

3. Step-by-Step Walkthrough of the Solution

Phase 1: Exploratory Data Analysis (EDA)

- **Univariate Analysis:** Revealed that `fare` is right-skewed, with a "long tail" of premium tickets >$600.
- **Bivariate Analysis:** Scatter plots of `nsmiles` vs. `fare` showed a weak correlation (~0.58), proving that distance alone does not dictate price.
- **Insight:** We discovered "Hub Premiums"—flights originating from major monopoly hubs (like Charlotte or Dallas) were consistently more expensive per mile than competitive routes.

Phase 2: Feature Engineering & Selection

- **The "Nuclear" Experiment:** We attempted Log-Transforming the target variable (`log_fare`). While mathematically sound, it led to overfitting on this specific dataset, dropping real-world accuracy to ~54%.
- **The "Recovery" & Breakthrough:** We reverted to predicting raw fare but introduced **Target Encoding**. By replacing city names (e.g., "Chicago") with their *historical average price*, we gave the model the spatial context it lacked. This was the turning point that pushed accuracy past 80%.

Phase 3: Hyperparameter Tuning

We utilized `RandomizedSearchCV` to optimize the XGBoost hyperparameters.

- **Winning Parameters:** `n_estimators=1000` , `learning_rate=0.01` , `max_depth=10` .
- **Logic:** A high number of trees with a very slow learning rate allowed the model to learn complex, subtle patterns without overfitting.

4. Model Evaluation

Champion Model: XGBoost Regressor

Metric	Value	Interpretation
R-Squared (R^2)	**82.34%**	The model explains 82.3% of the variance in ticket prices.
MAE (Mean Absolute Error)	**$19.36**	On average, our prediction is off by only $19.

RMSE (Root Mean Sq. Error)	$28.12	Penalizes large errors; consistent with the MAE, indicating few massive outliers.

Robustness Check: We validated the model using **3-Fold Cross-Validation**. The consistency of scores across different data splits confirms that the model is stable and not memorizing specific routes.

5. Comparison to Benchmark

Model	R^2 Score	MAE	Status
Linear Regression (Baseline)	58.00%	$30.50	Failed to capture non-linearity.
Random Forest	76.00%	$24.10	Good, but struggled with extremes.
XGBoost (Initial)	68.00%	$31.12	Underfit before Target Encoding.
XGBoost (Final Champion)	**82.34%**	**$19.36**	**Success** (41% improvement over baseline).

Verdict: The final solution significantly outperformed the baseline. The jump from 58% to 82% validates the hypothesis that airfare pricing is **highly non-linear** and heavily dependent on specific route characteristics rather than just distance.

6. Visualization(s)

(Include the following charts in your final document)

1. **Feature Importance Plot:** Shows `route_id_avg_price` and `monopoly_intensity` as the top drivers, visually proving our business hypothesis.
2. **Actual vs. Predicted Scatter Plot:** Shows the data points tightly hugging the diagonal line (perfect fit), a massive visual improvement over the scattered Baseline plot.
3. **30-Day Trend Graph (From App):** Displays the forecasted price movement, demonstrating the time-series capability.

7. Implications

Business Value:

- **For Travelers:** The "Fair Price" estimate empowers users to identify overpriced tickets. If the model predicts $300 and the airline asks $500, the user knows to wait or switch carriers.
- **For New Airlines:** The "Dominance Gap" analysis helps new entrants identify routes where incumbents are overcharging, highlighting opportunities for disruption.

Recommendation: We recommend stakeholders use the **"Deal Rating"** (Green/Red badge) generated by the app. This simplifies the complex $19.36 error margin into a binary "Buy/Don't Buy" signal, which is actionable for the average consumer.

8. Limitations

1. **Data Freshness:** The core dataset (Table 1a) is quarterly. Real-time pricing (e.g., "Price changed 5 mins ago") requires a live GDS API connection, which was out of scope.
2. **International Scope:** The model is currently trained only on US Domestic flights. International routes involves currency and geopolitical variables not yet modeled.
3. **Ancillary Fees:** The predicted fare is the "Base Fare." It does not account for baggage fees or seat selection costs, which are increasingly significant.

9. Closing Reflections

Learnings:

- **Feature Engineering > Algorithms:** We spent weeks tuning XGBoost for a 2% gain, but **Target Encoding** gave us a 14% gain in one day. Data quality and preparation trump model complexity.
- **Deployment Matters:** Building the Flask app forced us to handle "Real World" issues like missing inputs and user interface design, transforming a theoretical project into a tangible product.

Future Strategy: If we were to restart, we would implement **Target Encoding** immediately during the baseline phase. We would also prioritize integrating a live API (like Amadeus) earlier to enable real-time "Book Now" functionality.